

VAR Granger Causality/Block Exogeneity Wald Tests

Date: 04/26/26 Time: 17:06

Sample: 5/04/2016 3/30/2026

Included observations: 2453

Dependent variable: BRENT_RETURN

Excluded	Chi-sq	df	Prob.
BRENT_VOL	6.137210	4	0.1891
BRENT_VOLUME_CH	9.348714	4	0.0530
WTI_RETURN	1.882759	4	0.7573
WTI_VOL	9.624231	4	0.0473
WTI_VOLUME_CHAN	1.214189	4	0.8758
All	29.58473	20	0.0769

Dependent variable: BRENT_VOL

Excluded	Chi-sq	df	Prob.
BRENT_RETURN	14.36263	4	0.0062
BRENT_VOLUME_CH	14.93754	4	0.0048
WTI_RETURN	12.29691	4	0.0153
WTI_VOL	40.44949	4	0.0000
WTI_VOLUME_CHAN	2.719410	4	0.6058
All	82.43537	20	0.0000

Dependent variable: BRENT_VOLUME_CHANGE

Excluded	Chi-sq	df	Prob.
BRENT_RETURN	5.603611	4	0.2308
BRENT_VOL	5.515846	4	0.2383
WTI_RETURN	5.004541	4	0.2868
WTI_VOL	3.810405	4	0.4323
WTI_VOLUME_CHAN	13.29632	4	0.0099
All	33.11505	20	0.0328

Dependent variable: WTI_RETURN

Excluded	Chi-sq	df	Prob.
BRENT_RETURN	20.04000	4	0.0005
BRENT_VOL	5.653247	4	0.2266
BRENT_VOLUME_CH	5.725413	4	0.2206
WTI_VOL	7.676596	4	0.1042
WTI_VOLUME_CHAN	0.864300	4	0.9296
All	40.86805	20	0.0039

Dependent variable: WTI_VOL

Excluded	Chi-sq	df	Prob.
BRENT_RETURN	94.44959	4	0.0000
BRENT_VOL	150.2826	4	0.0000

BRENT_VOLUME_CH	2.716319	4	0.6064
WTI_RETURN	105.2420	4	0.0000
WTI_VOLUME_CHAN	2.084819	4	0.7202
All	258.3327	20	0.0000
Dependent variable: WTI_VOLUME_CHANGE			
Excluded	Chi-sq	df	Prob.
BRENT_RETURN	2.665011	4	0.6154
BRENT_VOL	1.766977	4	0.7785
BRENT_VOLUME_CH	3.830239	4	0.4295
WTI_RETURN	5.198229	4	0.2676
WTI_VOL	2.248409	4	0.6902
All	24.10453	20	0.2379